JAVASCRIPT CONDITIONAL STATEMENTS

20 Practice Tasks • Simple If-Else • If-Else-If-Else Ladder • Nested If-Else

Student Name: _____ **Date:** _____

Instructions

• Write a JavaScript program for each task.

• Use the specified conditional statement type for each section.

• You may use arithmetic, comparison, logical, and assignment operators wherever required.

• Do not use the ternary operator or switch statement.

• Try to solve the tasks independently. Show the output where appropriate.

SECTION 1: SIMPLE IF-ELSE (6 TASKS)

1. Even or Odd Number [Easy]

Write a program to accept a number and check whether it is even or odd.

Hint: Use the modulus operator %.

2. Voting Eligibility [Easy]

Write a program to check whether a person is eligible to vote. A person must be at least 18 years old.

3. Greater Number [Medium]

Accept two numbers and print the greater number. If both numbers are equal, print “Both numbers are equal”.

4. Discount Calculator [Medium]

Accept a product price. If the price is greater than or equal to ■5,000, give a 15% discount; otherwise, give a 5% discount. Display the discount amount and final price.

5. Number Validation [Medium]

Accept an integer and check whether it is a three-digit positive number. For example, 125 is valid, but 99, -125, and 1000 are invalid.

6. Loan Eligibility Check [Hard]

Accept a person's age, monthly salary, and credit score. Approve the loan only if the person is at least 21 years old, earns at least ■30,000 per month, and has a credit score of at least 700. Otherwise, reject the loan.

Hint: Use the logical AND operator &&.

SECTION 2: IF-ELSE-IF-ELSE LADDER (7 TASKS)

7. Positive, Negative, or Zero [Easy]

Accept a number and check whether it is positive, negative, or zero.

8. Student Grade Calculator [Easy]

Accept marks out of 100 and display the grade: 90–100: Grade A; 75–89: Grade B; 60–74: Grade C; 40–59: Grade D; below 40: Fail. If marks are below 0 or above 100, display “Invalid marks”.

9. Largest of Three Numbers [Medium]

Accept three numbers and print the largest number. If two or all three numbers share the maximum value, display an appropriate message.

10. Electricity Bill Calculator [Medium]

Accept the number of electricity units consumed and calculate the bill using these rates: up to 100 units: ■2 per unit; 101–200 units: ■3 per unit; 201–300 units: ■5 per unit; above 300 units: ■7 per unit. Apply the applicable rate to all units consumed. If units are negative, display “Invalid units”.

11. BMI Category Checker [Medium]

Accept weight in kilograms and height in meters. Calculate BMI = weight / (height × height). Display: below 18.5: Underweight; 18.5–24.9: Normal weight; 25–29.9: Overweight; 30 or above: Obese. Validate that both weight and height are greater than zero.

12. Employee Bonus Calculator [Hard]

Accept an employee's salary and years of experience. Experience of 10 years or more: 20% bonus; 5–9 years: 10%; 2–4 years: 5%; below 2 years: no bonus. If salary or experience is negative, display an invalid input message. Display the bonus and total salary after adding the bonus.

13. Smart Shopping Discount [Hard]

Accept the total shopping amount and customer type (regular or premium). Based on the total amount: ■10,000 or more: 20% discount; ■5,000–■9,999: 10%; below ■5,000: 2%. Premium customers receive an additional 5% discount on the original amount. Display the total discount and final bill. Reject negative amounts and invalid customer types.

SECTION 3: NESTED IF-ELSE (7 TASKS)

14. Login Authentication System [Easy]

Accept a username and password. First, check whether the username is correct. If correct, check whether the password is correct. If both are correct, check whether the account is active. If the account is active, display “Login Successful”; otherwise, display the appropriate error message. Use nested if-else statements for each check.

15. Exam Result and Scholarship [Medium]

Accept marks in three subjects: JavaScript, HTML, and CSS. First, check whether all marks are between 0 and 100. If valid, check whether the student has scored at least 35 in every subject. If the student passes, calculate the average. If the average is at least 90, award a 50% scholarship; otherwise, if the average is at least 75, award a 25% scholarship. Other passing students receive no scholarship. Display the result, average, and scholarship percentage.

16. ATM Withdrawal System [Medium]

Accept the account balance, withdrawal amount, and PIN. Check whether the PIN is correct. If correct, check whether the withdrawal amount is positive. If positive, check whether the amount is a multiple of ■100. If valid, check whether the balance is sufficient. If sufficient, deduct the amount from the balance and display the remaining balance. Display a suitable message for every failure.

17. Online Shopping Checkout [Medium]

Accept product price, quantity, stock availability, and customer membership (yes or no). Check whether the price and quantity are valid. If valid, check whether sufficient stock is available. If stock is available, calculate the total amount. If the total is at least ■5,000, apply a 10% discount. After that, if the customer is a member, apply an additional 5% discount to the amount remaining after the first discount. Display the final bill or an appropriate error message.

18. Job Eligibility System [Hard]

Accept age, qualification, years of experience, and a technical test score out of 100. Check whether the candidate is at least 18 years old. If eligible by age, check whether the qualification is a graduate degree. If qualified, check whether experience is at least 2 years. If experience is sufficient, check whether the technical score is at least 70. Display eligibility or the reason for rejection. Validate the test score and experience before evaluating eligibility.

19. Bank Loan Approval with Interest Rate [Hard]

Accept age, monthly income, credit score, and existing monthly EMI. Check whether age is between 21 and 60. If eligible, check whether monthly income is at least ■25,000. If income is sufficient, check whether the credit score is at least 700. If the score is sufficient, calculate the EMI-to-income ratio: (Existing EMI / Monthly Income) × 100. If the ratio is at most 30%, approve the loan at a 10% annual interest rate; otherwise, reject the application. Validate inputs and display the reason for rejection or the applicable interest rate.

20. Electric Vehicle Charging and Billing [Hard]

Accept battery percentage, battery capacity in kWh, desired charging percentage, and customer type (normal or premium). Check whether the current battery percentage is between 0 and 100. If valid, check whether the desired percentage is greater than the current percentage and at most 100. If valid, check whether battery capacity is greater than zero. Calculate energy required = capacity × (desired % – current %) / 100. If energy required is at most 10 kWh, charge ■8 per kWh; otherwise, charge ■12 per kWh. If the customer is premium, give a 10% discount. Display energy required, original bill, discount, and final bill. Reject invalid inputs and customer types.

Good luck! Practice carefully and test your code with different inputs.